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Crop Disease Detection and Classification System Using Computer Vision

Project Description:

This project focuses on developing a Computer Vision-based Crop Disease Detection and Classification System to identify whether a crop leaf is healthy or affected by a disease. Crop diseases can reduce crop quality and yield, while manual identification is time-consuming and may not always provide consistent results. The proposed system uses image processing and deep learning techniques to automatically analyse crop leaf images and classify them as healthy or diseased.

The project involves collecting and preparing crop leaf images, followed by image preprocessing, resizing, noise removal, background handling, segmentation, and feature analysis to improve the quality of the input images. Different approaches, including Traditional Computer Vision and Deep Learning, are considered and compared based on accuracy, adaptability, computational requirements, robustness, and deployment suitability. A suitable CNN/deep learning model is then trained and tested for crop disease classification. The performance of the system is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and graphical visualizations.

Tools Used: Python, OpenCV, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Google Colab, and GitHub.

Our Role in the Project: Our role is to design and implement the complete computer vision pipeline. We are responsible for dataset preparation, image preprocessing, model selection, model training and testing, performance evaluation, visualization of results, comparison of traditional and deep learning approaches, and project documentation. Through this project, we apply computer vision and machine learning concepts to develop a practical solution for early crop disease detection, which can help reduce crop losses and support sustainable agriculture.

1. Problem Understanding and Objectives

Agriculture is an important sector for food production, employment, and economic development. Crop diseases are one of the major causes of reduced yield and poor-quality agricultural products. Early identification of diseases is essential because delayed detection can allow infections to spread across large areas. Traditional disease identification mainly depends on manual inspection by farmers or agricultural experts. This process is time-consuming, subjective, and difficult to perform continuously over large farms. Therefore, a Computer Vision-based Crop Disease Detection and Classification System can provide an automated and efficient solution by analyzing crop leaf images and identifying whether the leaf is healthy or affected by a particular disease.

The proposed system accepts crop leaf images as input and performs image preprocessing, enhancement, segmentation, feature extraction, and disease classification. The system should handle variations such as illumination, background, leaf orientation, image quality, crop variety, and disease severity, while achieving a target accuracy of more than 95%. Two major approaches can be considered: traditional computer vision with machine learning and deep learning using CNN or transfer-learning models. Traditional methods depend heavily on manually designed colour, texture, and shape features, whereas deep learning automatically learns useful features from images. For a practical agricultural system, a suitable model should provide high accuracy while also considering computational cost, adaptability, scalability, and deployment requirements.

2. Proposed Computer Vision Pipeline

The proposed pipeline consists of several stages. First, crop leaf images are collected from a suitable dataset containing healthy and diseased leaves. The images are resized to a standard resolution so that they can be efficiently processed by the model. Image enhancement techniques such as noise removal, contrast enhancement, illumination correction, and filtering can be applied to improve image quality. Since field images may contain soil, other leaves, and complex backgrounds, segmentation techniques can be used to separate the leaf region from the background.

After segmentation, important visual characteristics of the leaf are analyzed. Traditional computer vision can extract colour features such as RGB, HSV, and colour histograms, texture features such as contrast, entropy, and Local Binary Patterns (LBP), and shape features such as area, perimeter, circularity, and contour properties. These features can then be provided to machine learning algorithms such as Support Vector Machine (SVM), Random Forest, or K-Nearest Neighbour (KNN) for classification. This approach is relatively simple and computationally efficient, but its performance depends strongly on the quality of manually selected features.

A deep learning approach can instead use a Convolutional Neural Network (CNN) to automatically learn disease-related patterns from leaf images. A transfer-learning model such as MobileNetV2, EfficientNet, or ResNet can be fine-tuned using the crop disease dataset. Transfer learning is particularly suitable when the available agricultural dataset is not extremely large because the model starts with knowledge learned from a large image dataset. Data augmentation techniques such as rotation, flipping, cropping, scaling, and brightness variation can improve robustness to different field conditions.

3. Traditional Computer Vision vs Deep Learning

Factor	**Traditional CV + ML**	**Deep Learning**
Feature extraction	Manually designed colour, texture and shape features	Automatically learned features
Accuracy	Good for controlled images	Generally higher for complex images
Data requirement	Relatively low	Usually requires more data
Adaptability	Limited when conditions change	Better adaptability with diverse training data
Computational cost	Low to moderate	Moderate to high
Training complexity	Simpler	More complex
Background/illumination variation	Can significantly affect performance	More robust when trained with varied images
Scalability	Requires feature engineering for new conditions	Easier to scale to multiple diseases/crops
Deployment	Suitable for low-resource devices	Lightweight models can support edge/mobile deployment
Suitability	Useful as a baseline/comparison	Recommended for the proposed system

For this application, deep learning with transfer learning is the preferred approach. A model such as EfficientNet or MobileNetV2 can provide a good balance between classification accuracy and computational requirements. EfficientNet can provide strong image classification performance, while MobileNetV2 is particularly attractive when the final system needs to operate on mobile phones, edge devices, or low-resource agricultural equipment. The final model should be selected based on experimental results rather than assuming that any particular model will automatically achieve 95% accuracy.

4. Algorithm / Pseudocode

BEGIN

1. Collect crop leaf image dataset
2. Divide dataset into training, validation and testing sets
3. Resize all images to a fixed image size
4. Apply preprocessing:
    - Noise removal
    - Illumination correction
    - Normalization
    - Background/leaf segmentation
5. Apply data augmentation to training images
6. Load a pretrained CNN/transfer-learning model
7. Replace the final classification layer with crop disease classes
8. Train the model using training images
9. Validate the model after each training epoch
10. Tune learning rate, batch size and number of epochs
11. Evaluate the final model using test images
12. Calculate:

- Accuracy

- Precision

- Recall

- F1-score

13. Generate confusion matrix and prediction visualizations

14. Compare results with traditional ML approach

15. Select the model based on accuracy, robustness,

 computational cost and deployment requirements

16. Predict disease for a new crop leaf image

17. Display predicted disease/healthy status and confidence score

END

5. Evaluation and Expected Results

The performance of the system should be evaluated using a separate testing dataset that is not used during model training. **Accuracy** measures the overall percentage of correctly classified images, while **precision** indicates how accurately the system identifies a particular disease. **Recall** measures the ability to detect actual diseased leaves, which is especially important in agriculture because missing an infected plant can allow the disease to spread. The **F1-score** provides a balance between precision and recall. A **confusion matrix** can be used to identify which disease classes are frequently confused with one another.

The system should also provide visualizations such as training and validation accuracy graphs, training and validation loss graphs, confusion matrices, and sample predictions showing the original leaf image along with the predicted class. A comparative experiment can be conducted using a traditional machine learning model and the selected deep learning model. The comparison should consider not only accuracy but also inference time, model size, training requirements, robustness, and deployment feasibility. The objective is to select a model capable of meeting the **>95% target accuracy**, provided that this target is supported by the test results.

6. Agricultural Relevance and SDG Mapping

The proposed system can support early crop disease detection, allowing farmers to identify infections before they spread extensively. Automated detection can reduce dependence on continuous manual inspection, save time, and help farmers take timely corrective measures. Early intervention can reduce crop losses, improve productivity, and support more efficient use of agricultural resources. When deployed through a mobile or edge-based application, farmers could capture a leaf image and receive an automated disease prediction.

The project directly supports SDG 2 – Zero Hunger through improved agricultural productivity, while the given SDG mapping particularly relates to SDG 9 – Industry, Innovation and Infrastructure, SDG 12 – Responsible Consumption and Production, and SDG 8 – Decent Work and Economic Growth. By applying artificial intelligence and computer vision to agriculture, the system promotes technological innovation, reduces unnecessary resource usage, and supports sustainable and productive farming practices. Overall, the project demonstrates how computer vision, machine learning, and deep learning can be integrated to create a practical smart-agriculture solution.

1. Problem Analysis and Pseudocode

Problem Understanding

The proposed Crop Disease Detection and Classification System Using Computer Vision aims to automatically identify whether a crop leaf is healthy or affected by a disease. Crop diseases can reduce agricultural productivity and cause significant economic losses when they are not detected at an early stage. Manual identification requires expert knowledge, is time-consuming, and may produce inconsistent results. The proposed system uses image processing and machine/deep learning techniques to analyze crop leaf images and classify their health condition automatically.

The system must handle practical variations such as different illumination conditions, complex backgrounds, leaf orientation, image quality, crop varieties, and different levels of disease severity. The main performance objective is to achieve more than 95% classification accuracy while maintaining good precision, recall, and F1-score. The system should also be computationally efficient enough for possible deployment in agricultural applications.

Functional Requirements

1. Accept crop leaf images as input from a dataset or camera/mobile device.

2. Resize and normalize images into a suitable format.
3. Remove noise and improve image quality using preprocessing techniques.
4. Segment the leaf from complex backgrounds when required.
5. Extract colour, texture, and shape features for traditional machine learning.
6. Train a CNN or transfer-learning model for automatic feature extraction and classification.
7. Classify images into healthy or corresponding disease categories.
8. Evaluate the model using accuracy, precision, recall, and F1-score.
9. Generate a confusion matrix and training/validation graphs.
10. Display the predicted disease class and confidence score for new images.

Comparison and Selected Approach

Two approaches are considered. The first is Traditional Computer Vision with Machine Learning, where colour, texture, and shape features are manually extracted and provided to classifiers such as SVM, Random Forest, or KNN. This approach requires less computational power and can work with smaller datasets, but its performance can decrease when there are changes in lighting, background, disease appearance, and crop variety.

The second approach is Deep Learning, where CNN or transfer-learning models automatically learn important visual features from leaf images. Deep learning generally provides better adaptability and scalability for complex image classification problems. Therefore, the proposed system selects a transfer-learning-based CNN model such as EfficientNet or MobileNetV2. Transfer learning reduces training requirements and provides a good balance between accuracy and computational efficiency. The final model should be selected based on experimental evaluation and its ability to satisfy the >95% target.

Pseudocode

BEGIN

Load crop leaf image dataset

Divide dataset into:

Training set

Validation set

Testing set

FOR each image:

Resize image

Remove noise

Correct illumination

Normalize pixel values

Segment leaf region if required

END FOR

Apply data augmentation to training images

Load pretrained CNN model

Modify final classification layer

according to disease classes

Train model using training dataset

Validate model during training

Tune model parameters if required

Evaluate model using testing dataset

Calculate:

Accuracy

Precision

Recall

F1-score

Generate confusion matrix

Generate accuracy and loss graphs

Compare deep learning results

with traditional ML results

Select best-performing model

FOR a new input leaf image:

Preprocess image

Predict disease class

Generate confidence score

Display result

END FOR

END

Github Link:

Kanisghayazhini.C(https://github.com/kanisgha/ITA0501-Computer-Vision/tree/main/Crop%20Disease%20Detection)

SathwikaReddy(https://github.com/192525274/ITA0501-Computer-Vision/tree/main/Crop_Disease_Detection)